Unit: 6 Unit 8: Quadratics **Name:** ______________
Topic: Vertex Form and Intercept Form **Date:** ___________

Advanced (Habanero)

Q1. What is the vertex form of a quadratic function?
(A) $y = a(x-h)^2 + k$
(B) $y = a(x+h)^2 - k$
(C) $y = ax^2 + bx + c$
(D) $y = a(x-h)(x-k)$
(E) $y = ax^2 - bx - c$

Q2. Which of the following is the intercept form of $y = x^2 - 4x - 5$?
(A) $y = (x-2)(x+1)$
(B) $y = (x+2)(x-1)$
(C) $y = (x-1)(x+5)$
(D) $y = (x+1)(x-5)$
(E) $y = (x-5)(x+1)$

Q3. What is the value of a in $y = a(x-2)^2 + 1$ if the graph passes through $(0,5)$?
(A) 2
(B) 1
(C) -1
(D) -2
(E) 4

Q4. What are the x-intercepts of $y = (x-1)(x+3)$?
(A) $x = -1, x = 3$
(B) $x = 1, x = -3$
(C) $x = -1, x = -3$
(D) $x = 1, x = 3$
(E) $x = 0, x = 2$

Q5. What is the vertex of $y = x^2 - 4x + 3$?
(A) $(2, -1)$
(B) $(-2, 1)$
(C) $(0, 3)$
(D) $(2, 3)$
(E) $(-2, -1)$

Q6. Which equation represents a quadratic function with its vertex at $(0,0)$?
(A) $y = x^2$
(B) $y = -x^2$
(C) $y = x^2 - 1$
(D) $y = -x^2 + 1$
(E) $y = x^2 + 1$

Q7. What is the value of k in $y = (x-2)^2 + k$ if the graph passes through $(1,2)$?
(A) -2
(B) -1
(C) 0
(D) 1
(E) 2

Q8. What is the equation of the axis of symmetry of $y = x^2 - 4x - 3$?
(A) $x = -2$
(B) $x = 2$
(C) $x = -1$
(D) $x = 1$
(E) $x = 0$

Open Ended Questions (FRQ)

Q9. Find the vertex form of $y = x^2 + 2x - 6$ and identify the vertex.

Q10. Write the equation of a quadratic function with its vertex at $(-2, 3)$ and x-intercepts at $(-4, 0)$ and $(0, 0)$.

Chef's Tips

- Tip 1: Encourage students to use graphing tools to visualize the quadratic functions.
- Tip 2: Emphasize the importance of identifying the parameters in vertex and intercept forms.
- Tip 3: Remind students to check their work by plugging in values into the equations.